

R3M: A Universal Visual Representation for Robot Manipulation

[CoRL, 2022]

Introduction

End-to-End learning refers to a training approach where the entire process—from input to output—is learned by a single neural network or model.

When applied to robotics,

this means mapping sensor inputs (e.g., camera images, LiDAR data) directly to action outputs (e.g., steering, speed control) **without explicitly separating intermediate steps** like perception, planning, and control

In short, the model **"learns actions directly from data."**

Category	Traditional Robotic System	End-to-End Learning
Architecture	Perception → Planning → Control (Modular)	Unified in a single neural network
Required Knowledge	Extensive domain knowledge	Large-scale data
Advantages	Interpretability, Stability	Flexibility, Simplification
Disadvantages	Requires complex design	Training stability issues, Black-box nature

Introduction

Conventional Robotic Manipulation Learning: End-to-End Approach

The manipulation model is trained using data collected by repeatedly performing the same task within a specific domain—i.e., a particular robot manipulating a particular object.

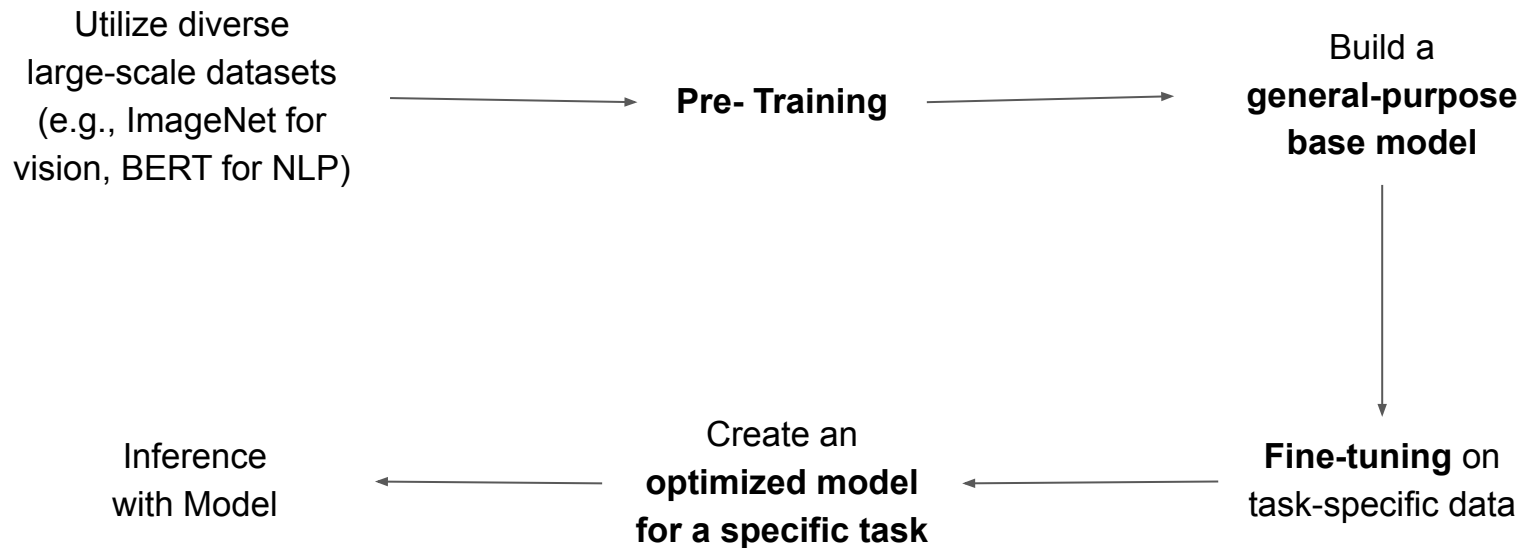
Limitations:

- Requires **large-scale data collection** for each specific manipulation task
- **Generalization is difficult**, as models tend to be **task-specific** and not agnostic to manipulation types



Introduction

Shift in Vision/NLP: Pretrained Representations



Introduction

However, **robotics still lacks general-purpose pretrained representations** due to several key challenges:

1. **High Data Collection Costs**

Interaction data in the physical world is expensive to collect and difficult to annotate.

2. **Lack of Task and Environment Diversity**

Existing robotic datasets are relatively small in scale and limited to specific, constrained environments.

3. **Difficulty in Learning Generalized Representations**

There is a significant lack of data diversity necessary for training representations that can generalize across various tasks and settings.

Introduction

Alternative: Human-Centric Videos

(large-scale videos of people interacting with the real world — e.g., cooking, cleaning, moving)

- Despite differences between humans and robots, such data has proven useful in areas like **reward function learning in reinforcement learning**.
- Similar to vision and NLP domains, **pretrained representations** learned from human activity videos **may be reusable** in robotics.

Human daily-life data can serve as **pretraining data** to help robots learn more effectively.

Introduction

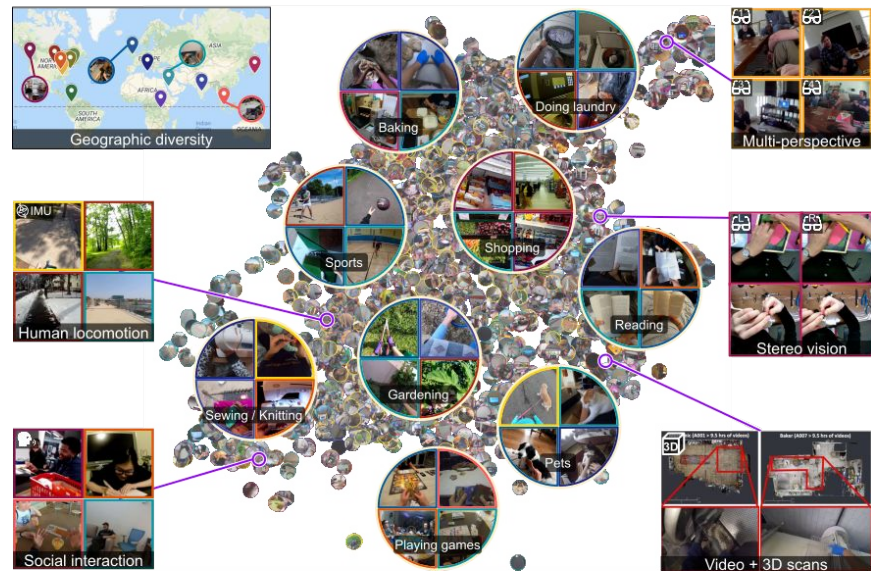
Ego4D

Developed by **Meta AI**, Ego4D is the **world's largest first-person (egocentric) video dataset**, designed to capture everyday human experiences from the user's point of view.

- Participants wear **head-mounted cameras** to record their daily lives
- Unlike third-person datasets, Ego4D focuses on **human gaze, attention, and interaction**

Large-Scale & Multinational Dataset

- Over **3,600 hours** of video
- Collected from **74+ participants** across **9 countries**
- Includes **diverse cultural contexts** and real-world environments



Introduction

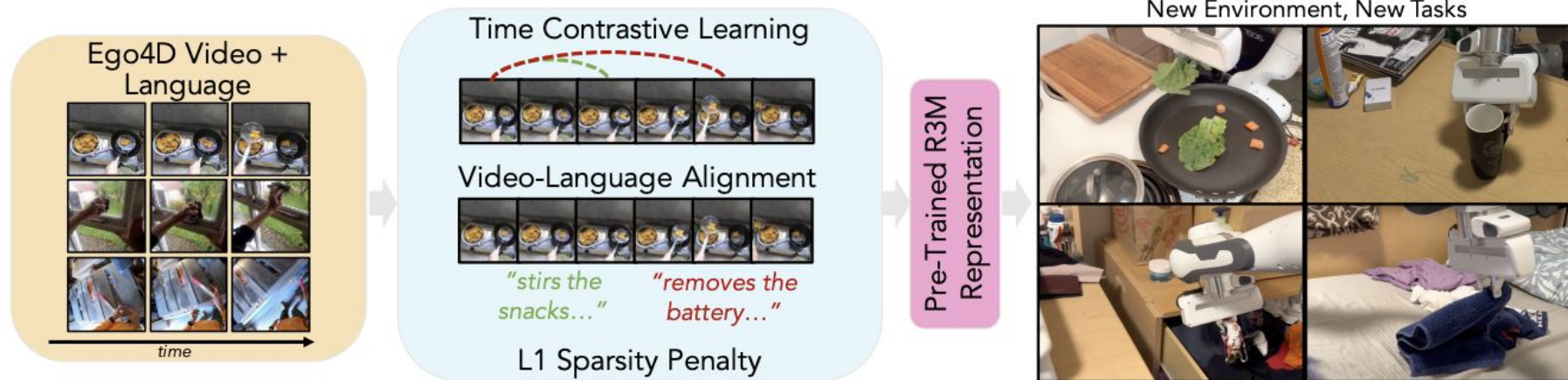
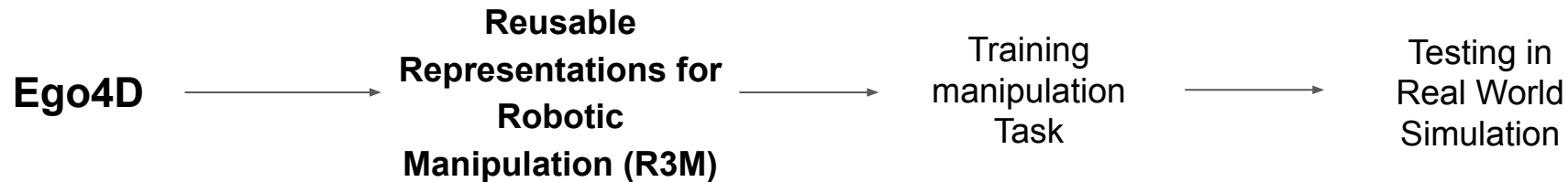
Three Key Requirements for Vision-Based Robotic Manipulation Representations

1. **Capture Physical Interaction and Temporal Dynamics**
→ The representation must reflect how a scene evolves over time, enabling the robot to understand and respond to state changes in a sequential manner.
2. **Embed Semantic Priors of Task-Relevant Elements**
→ The representation should focus on semantically important objects and their relationships—those that are essential for completing the task.
3. **Be Compact and Filter Out Irrelevant Information**
→ Background and other non-essential features should be excluded, resulting in a sparse and concise representation.

To meet these requirements, the following techniques are employed:

- **Time-Contrastive Learning (TCL):**
Learns scene dynamics by distinguishing changes over time, enabling understanding of temporal state transitions.
- **Video-Language Alignment:**
Aligns video frames with natural language descriptions to encode semantic priors—helping the model identify meaningful objects and interactions.
- **L1 and L2 Regularization:**
Enforces compactness and sparsity in representations by suppressing unimportant features.

Introduction

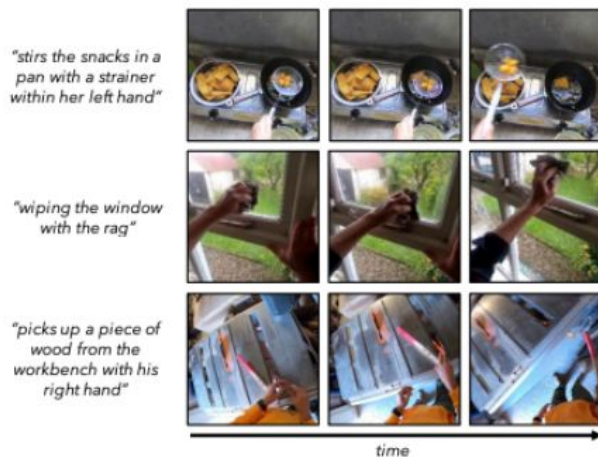


Model

Objective

Leverage diverse human video datasets to **pretrain a single, reusable visual representation** that is particularly useful for **robot motor control**, with a focus on **manipulation tasks**.

Ultimately, the goal is to enable **efficient downstream learning**—even in **previously unseen environments and tasks**.



Downstream Task: An agent must **sequentially choose actions** based on **image observations I** , learning how to interact with the environment effectively using the pretrained visual representation.

Model

Assuming we have a dataset **D** consisting of **N videos**, where:

- Each video consists of RGB frames:
 $D = [I_0, I_1, \dots, I_T]$
- Each video is paired with a **natural language description** **I**, which describes the task occurring in the video.

We define an **image encoder** **F_ϕ** , which transforms a single image into a **deterministic, continuous embedding**:

$$\mathbf{z} = F_\phi(I)$$

where **z** is the **state representation**.

To make the **state representation** **$\mathbf{z} = F_\phi(I)$** a viable substitute for raw images in robotic learning, the representation must satisfy key requirements to ensure broad **utility and generalization** across tasks and environments.

*The encoder **F_ϕ** must be **reusable across multiple downstream policy learning tasks**.

Model

Requirements for an Effective Representation $F\phi$:

1. **Trained on Sufficiently Diverse Data**

→ To promote **generalization**, the encoder must be exposed to a wide range of tasks, objects, and environments during pretraining.

2. **Provide Relevant Signals for Robotic Manipulation**

→ The representation should emphasize features that are critical for **robotic control and decision-making**, such as object states, spatial relations, and motion cues.

To meet these conditions, use **video data showing humans interacting with the real world**, especially data that:

- Captures rich, purposeful **interactions** similar to those a robot must perform
- Contains **first-person perspectives**, aligning with robotic viewpoints
- Reflects **scene dynamics** and **semantic understanding** required in manipulation tasks

→ **Ego4D** is an ideal dataset for this purpose, offering large-scale, egocentric recordings of human activities across diverse environments and cultures.

Model

"stirs the snacks in a pan with a strainer within her left hand"



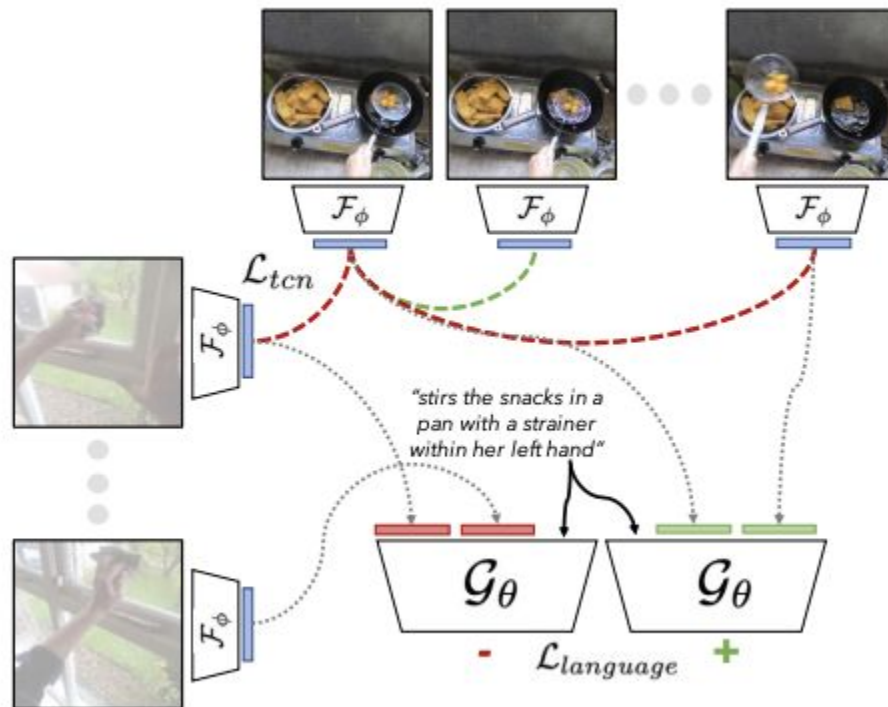
"wiping the window with the rag"



"picks up a piece of wood from the workbench with his right hand"



time



Final Form of R3M Loss Function

$$\mathcal{L}(\phi, \theta) = \mathbb{E}_{I_{0,i,j,k}^{1:B} \sim \mathcal{D}} [\lambda_1 \mathcal{L}_{tcn} + \lambda_2 \mathcal{L}_{language} \\ + \lambda_3 ||\mathcal{F}_\phi(I_i)||_1 + \lambda_4 ||\mathcal{F}_\phi(I_i)||_2]$$

Model

Time-Contrastive Learning (TCL)

- **Purpose:**
Train encoder $\mathbf{F}\phi$ to capture features important for **physical interaction** and **sequential decision-making**
- **Objective Function:**
First training objective: **Time-Contrastive Loss (TCL)**
- **Sampling Method:**
Sample frame sequences $(\mathbf{l}_i, \mathbf{l}_{\square} > \mathbf{l}_i, \mathbf{l}_{\square} > \mathbf{l}_{\square})$ from each video in a batch $\mathbf{B}$
- **Training Behavior:**
 - **Temporally close frames** $\rightarrow$ similar embeddings
 - **Distant frames or different videos** $\rightarrow$ dissimilar embeddings

$$S(P, Q) =$$

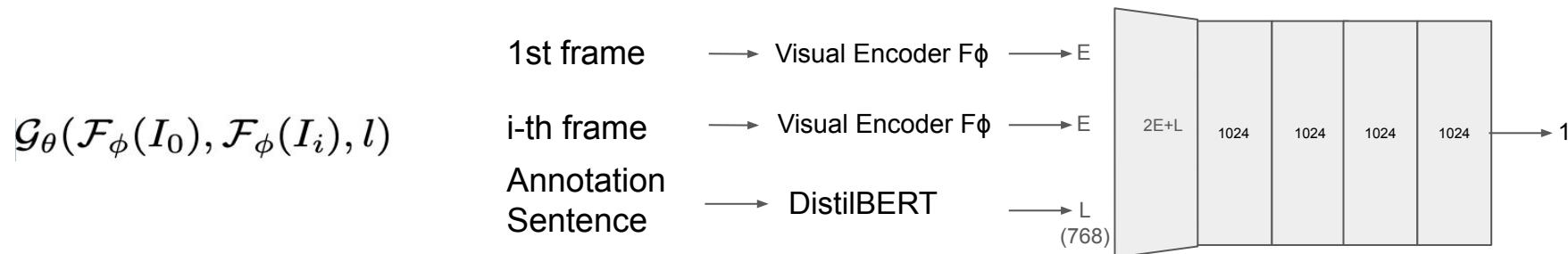
$$\sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \dots + (p_n - q_n)^2} = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

$$\mathcal{L}_{tcn} = - \sum_{b \in B} \log \frac{e^{\mathcal{S}(z_i^b, z_j^b)}}{e^{\mathcal{S}(z_i^b, z_j^b)} + e^{\mathcal{S}(z_i^b, z_k^b)} + e^{\mathcal{S}(z_i^b, z_i^{\neq b})}}$$

Model

Video-Language Alignment

- **Purpose:**
Capture **semantically meaningful features**
→ Train a module to predict language from embeddings generated by F_ϕ
- **G_θ Training Goal:**
 - Score should **increase** as the video progresses toward fulfilling the language description
 - **Correct video-language pairs** should receive **higher scores** than incorrect ones
- **Scoring Behavior:**
Predict a **low score** when the scene transition **does not match** the given language instruction



Model

Annotation
Sentence

$$\mathcal{G}_{\theta}(\mathcal{F}_{\phi}(I_0), \mathcal{F}_{\phi}(I_i), l)$$

1st frame

i-th frame

$$\mathcal{L}_{language} = - \sum_{b \in B} \log \frac{e^{\mathcal{G}_{\theta}(z_0^b, z_{j>i}^b, l^b)}}{e^{\mathcal{G}_{\theta}(z_0^b, z_{j>i}^b, l^b)} + e^{\mathcal{G}_{\theta}(z_0^b, z_i^b, l^b)} + e^{\mathcal{G}_{\theta}(z_0^{\neq b}, z_{j>i}^{\neq b}, l^b)}}$$

Model

Regularization

- **Assumption:**
Sparse and compact representations are beneficial for manipulation control.
→ Especially effective in low-data imitation learning settings.
- **Imitation Learning Issue:**
A common failure is **state-distribution shift** — when the policy deviates from the expert's state trajectory.
- **Solution:**
Reduce the **effective dimensionality** of the state representation
→ Helps maintain alignment with expert behavior.
- **Implementation:**
Apply **L1 and L2 regularization** to enforce compact, sparse representations.

$$||\mathcal{F}_\phi(I_i)||_1$$

$$||\mathcal{F}_\phi(I_i)||_2$$

Model

Final Form of R3M Loss Function

$$\mathcal{L}(\phi, \theta) = \mathbb{E}_{I_{0,i,j,k}^{1:B} \sim \mathcal{D}} [\lambda_1 \mathcal{L}_{tcn} + \lambda_2 \mathcal{L}_{language} \\ + \lambda_3 ||\mathcal{F}_{\phi}(I_i)||_1 + \lambda_4 ||\mathcal{F}_{\phi}(I_i)||_2]$$

Simple objective function → Can be implemented with any encoding architecture

Experiment

Experiment Objectives

1. **Data Efficiency in Unseen Scenarios**

- Does R3M enable more data-efficient imitation learning compared to baseline methods (e.g., from-scratch learning or other pretrained features)?

2. **Ablation Study**

- Evaluate the contribution of each component in R3M's training objective by selectively removing them.

3. **Real-World Generalization**

- Can R3M support efficient policy learning in visually complex, real home environments using real robots?

Experiment

4.1 Imitation Learning Evaluation Framework

Focus on evaluation of a frozen pretrained visual representation module as input to policy learning via behavior cloning

Pretrained visual representation $F\phi$,

State Representation : a concatenation of the visual embedding $z_t = F\phi(I_t)$ + robot proprioceptive (e.g. joint positions and velocities) reading p_t .

a standard behavior cloning loss
 $\|a_t - \pi([z_t, p_t])\|_2^2$

—————→ The policy, π

Training Details : Trained for 20,000 steps; online evaluation every 1,000 steps; final result is based on the policy with the highest success rate

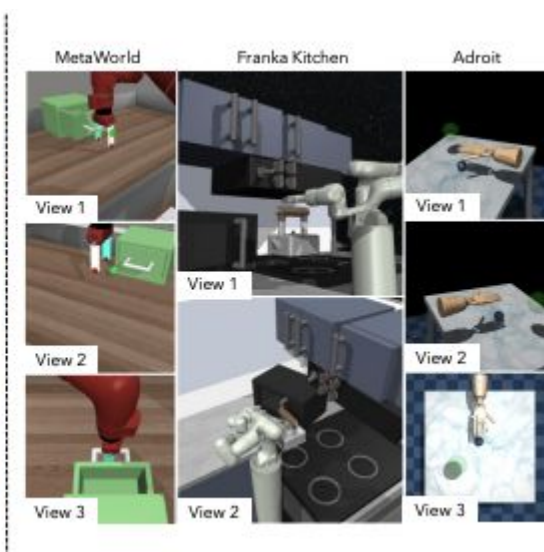
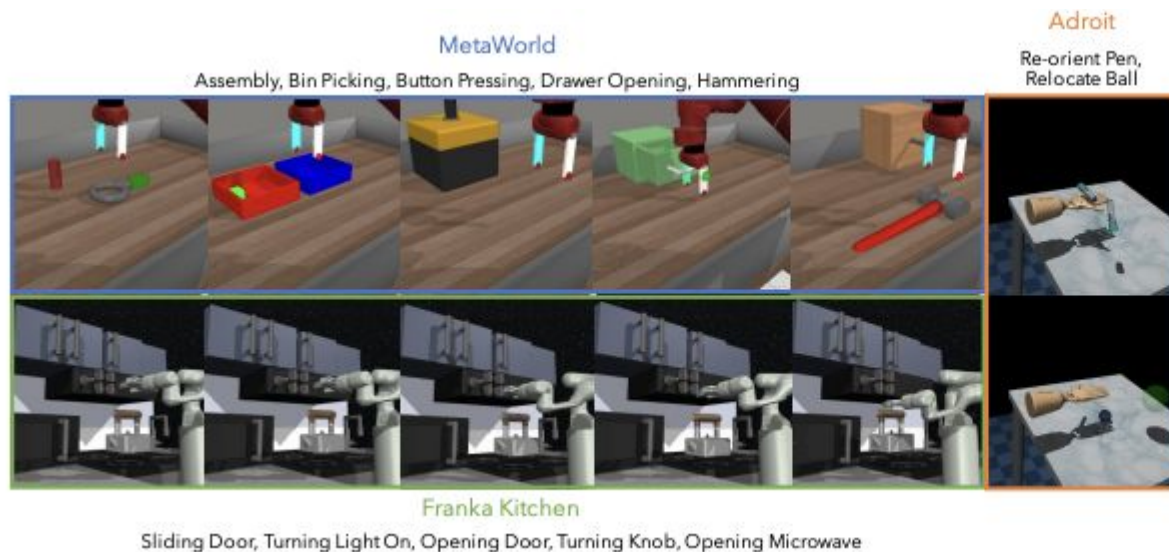
Policy Network : The policy π is implemented as a two-layer MLP with BatchNorm applied to the input

Experiments

4.2 Simulation Environments

Evaluation Protocol

- For each **visual representation** and **task**, experiments were run with:
 - 3 random seeds
 - 2–3 different camera viewpoints
- The performance of **R3M** and **baseline models** was compared using the **average success rate across all environments**.



Experiments

4.3 Exp. 1: Does R3M enable efficient imitation on unseen environments and tasks?

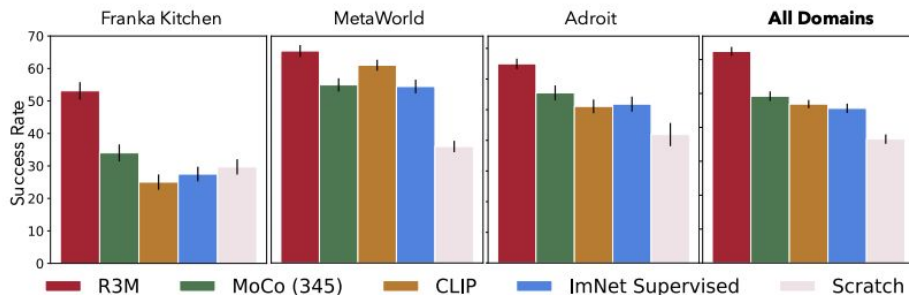


Figure 4: **Data Efficient Imitation Learning in Unseen Environments/Tasks.** We report the success rates of downstream imitation learning with standard error bars. We observe that across 12 tasks R3M outperforms baselines like MoCo (345) (PVR), CLIP, Supervised ImageNet features, and training from scratch.

Conclusion

Pretrained visual representations are essential for **data-efficient imitation learning**.

R3M is particularly effective for learning **robotic manipulation tasks**.

Measure the imitation learning success rate using various pretrained visual encoders.

Key Results (Figure 4)

- **R3M** achieved **~62% success rate** with very limited data
- Outperformed training from scratch by **over 20%**
- Averaged **10%+ higher performance** across 12 tasks compared to other pretrained features
- R3M was the **top performer** in **11 out of 12 tasks** across **3 different environments**

Model Comparison

- **CLIP**: Strong performance in MetaWorld
- **MoCo (ResNet-34/ResNet-45)**: Better on Franka and Adroit tasks
- **From-scratch learning**: Performed poorly under low-data conditions

Experiments

4.4 Exp. 2: Which components of R3M are important?

Environment	<i>Supervised</i>				<i>Self-Supervised</i>
	R3M	R3M(-Aug)	R3M(-L1)	R3M(-Lang)	R3M(-Lang)
Franka Kitchen	53.1 $\pm 2.7\%$	51.1 $\pm 2.7\%$	46.7 $\pm 2.7\%$		47.2 $\pm 2.9\%$
MetaWorld	69.2 $\pm 2.0\%$	68.9 $\pm 2.1\%$	65.0 $\pm 2.4\%$		67.0 $\pm 2.0\%$
Adroit	65.0 $\pm 1.7\%$	61.3 $\pm 2.1\%$	66.5 $\pm 1.6\%$		45.6 $\pm 3.3\%$
All Domains	62.4 $\pm 1.3\%$	60.4 $\pm 1.4\%$	59.4 $\pm 1.5\%$		53.2 $\pm 1.5\%$

Table 1: **Ablating Components of R3M.** We see report success rate of downstream imitation learning on variants of R3M. We observe that on average, removing the L1 penalty have a negative impact, particularly on the Franka Kitchen and MetaWorld environments. Lastly, removing language grounding has the most significant drop in performance, particularly on the Adroit tasks.

$$\mathcal{L}_{\text{language}} = - \sum_{b \in B} \log \frac{e^{\mathcal{G}_{\theta}(z_0^b, z_j^b, i, l^b)}}{e^{\mathcal{G}_{\theta}(z_0^b, z_j^b, i, l^b)} + e^{\mathcal{G}_{\theta}(z_0^b, z_i^b, l^b)} + e^{\mathcal{G}_{\theta}(z_0^{\neq b}, z_j^{\neq b}, i, l^b)}}$$

Key Insight

Even without language alignment (**fully self-supervised version of R3M**),

R3M **still outperforms** CLIP and MoCo (345).

Ablation Study: R3M Variants

Experimental Variants

- **R3M (-Aug):** No crop augmentations
- **R3M (-L1):** No L1 regularization
- **R3M (-Lang):** No video-language alignment loss

Results Summary (Table 1)

- **Removing crop augmentations or L1 regularization** $\rightarrow$ ~2% average performance drop
- **Effect of L1 regularization varies by environment:**
 - **MetaWorld / Franka:** L1 improves performance
 - **Adroit:** L1 removal improves performance
 - Likely due to reduced state distribution shift (more demonstrations available)
- **Removing video-language alignment loss** causes the **largest performance drop**
 - Most significant impact in **Adroit environment**
 - Highlights the importance of **semantic feature grounding**

Experiment - Can R3M Enable Data-Efficient Learning in the Real World?

4.5 Exp. 3: Does R3M enable data efficient learning in real world environments?

Success out of 10 trials	R3M	CLIP
Closing Drawer	80%	70%
Putting Mask in Dresser	30%	10%
Putting Lettuce in Pan	60%	0%
Pushing Mug to Goal	70%	40%
Folding Towel	40%	0%
Average	56%	24%

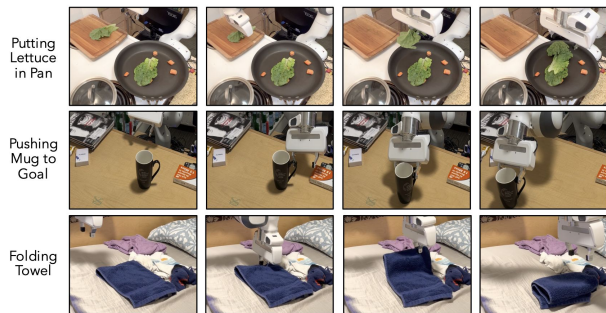
Table 3: **Real World Success Rates.** R3M outperforms CLIP on the challenging real world manipulation tasks.

Tasks Performed

- Closing a drawer
- Picking up a randomly placed mask and putting it in a drawer
- Placing lettuce from a cutting board into a pan
- Pushing a cup to a target location
- Folding a towel
- **Performance Results (Table 3, Figure 5)**
 - On the **easiest task** (closing a drawer), **R3M and CLIP** perform similarly
 - On the **other 4 tasks**, **R3M consistently outperforms CLIP**, showing about **2× higher success rates** and more stable performance

Real-World Setup

- Used a **Franka Emika Panda** robot
- Deployed in a **real apartment** (graduate student's home)
- Only **20 demonstrations per task**
- Trained using **only visual input** from the **pretrained R3M model**



Limitations

Limit 1: Narrow Evaluation Scope

- Current evaluation is limited to **imitation learning**, specifically **behavior cloning** with only a few demonstration samples.
- A pretrained representation effective for imitation learning is **not guaranteed** to perform well in **reinforcement learning (RL)**.

Limit 2: Limited to Single-Frame Representations

- The current R3M model provides **only single-frame state embeddings**.
- To enhance applicability, it should be extended to support:
 - **Reward learning**
 - **Task specification**
 - **Temporal or multi-frame understanding**